Experiment 15: Family Knowledge Representation

Aim:

To represent family relationships as facts and rules and to infer new relationships such as father, mother, sibling, brother, sister and grandparent using logical rules.

Algorithm:

1. Store the basic facts: the set of male persons, female persons and parent-child pairs.
2. Define rules: father(X,Y) if X is a parent of Y and X is male; mother(X,Y) if X is a parent of Y and X is female.
3. Define sibling(X,Y) if X and Y are different persons sharing a common parent; brother and sister add the gender condition.
4. Define grandparent(X,Y) if X is a parent of some Z and Z is a parent of Y.
5. Query the knowledge base and infer the relationships using the rules.

Code:

```
# Facts
male = {"john", "david", "mike", "tom"}
female = {"mary", "susan", "anna"}
parent_of = [
    ("john", "david"), ("mary", "david"),
    ("john", "susan"), ("mary", "susan"),
    ("david", "tom"), ("anna", "tom"),
]

# Rules
def parent(x, y):
    return (x, y) in parent_of

def father(x, y):
    return parent(x, y) and x in male

def mother(x, y):
    return parent(x, y) and x in female

def sibling(x, y):
    people = {p for pair in parent_of for p in pair}
    return x != y and any(parent(p, x) and parent(p, y) for p in people)

def brother(x, y):
    return sibling(x, y) and x in male

def sister(x, y):
    return sibling(x, y) and x in female

def grandparent(x, y):
    people = {p for pair in parent_of for p in pair}
    return any(parent(x, z) and parent(z, y) for z in people)

def grandfather(x, y):
    return grandparent(x, y) and x in male

people = sorted({p for pair in parent_of for p in pair})

print("Is john the father of david?", father("john", "david"))
print("Is mary the mother of susan?", mother("mary", "susan"))
print("Is mary the father of susan?", father("mary", "susan"))
print("Is david a brother of susan?", brother("david", "susan"))
print("Is susan a sister of david?", sister("susan", "david"))
print("Is john a grandfather of tom?", grandfather("john", "tom"))
print("Grandparents of tom:", [p for p in people if grandparent(p, "tom")])
print("Children of john:", [p for p in people if parent("john", p)])
print("Siblings of david:", [p for p in people if sibling(p, "david")])
```

Sample Output:

Output

```
Is john the father of david? True
Is mary the mother of susan? True
Is mary the father of susan? False
Is david a brother of susan? True
Is susan a sister of david? True
Is john a grandfather of tom? True
Grandparents of tom: ['john', 'mary']
Children of john: ['david', 'susan']
Siblings of david: ['susan']
```

Result:

The knowledge base inferred relationships that were never stored directly. It concluded that john is the father of david, david and susan are siblings, and john and mary are the grandparents of tom.